

8

In BitTorrent, if Alice has a subset of chunks and knows which chunks her neighbors have, then Alice requests small size chunks first from her neighbors. In this manner she gets chunks more quickly and will have a lot of chunks which helps her to be one of the top 4 for other peers 

☐ True

☒ False

 **Will be reviewed**

9

Which services can be granted and can't be granted by TCP when it uses flow control and congestion control? (Justify your answer) 

TCP only grants the reliability but does not grant security or time or throughput

 **Will be reviewed**

10

Why is a Web cache considered both a server and a client at the same time?



It sends the requested messages to the clients so it acts like a server and it requests the pages from the main server to cache it so it acts as a client

 **Will be reviewed**

11

Why do Mail servers use SMTP handshaking in application layer although there is a TCP handshaking in the transport layer? 

to check if the email is sender is not blocked and ok to send
and to check if the receiver mail is in the mail server

 **Will be reviewed**

12

Question 

☐ True

☒ False

 **Will be reviewed**

13

Dial-up and DSL are both dedicated access technology, where HFC and FTTH(PON) are completely shared along the path. 

☐ True

☒ False

📝 Will be reviewed

14

Assume that a file of size F is to be distributed to N clients in client-server architecture. If the upload rate of the server's access link is us , then the time to distribute the file to N clients is equal to NF/us 

☐ True

☒ False

📝 Will be reviewed

15

Consider an institution has an access line of 15 Mbps. The access line services 20 requests per second. Suppose that the average size of a request is 1Mbits. Consider that the Internet delay is 2 sec; LAN delay is 50 msec. What is the average delay experienced by requests when using proxy server with hit rate =20%? (justify your answer). 

hit rate is 20% then the file size that will be transmit on the link is $0.8 \times 20 \times (1 \times 10^6) = 16 \times 10^6$ bit

the traffic delay = $(16 \times 10^6) / (15 \times 10^6) = 1.06$ which is more than 1 the the delay will be infinity

📝 Will be reviewed

16

Question 

☐ True

☒ False

📝 Will be reviewed

17

UDP socket is fully identified by a source port number, and a destination port number while TCP socket is fully identified by a source port number, a destination port number, a source IP address and a destination IP address 

☒ True

☐ False

📝 Will be reviewed

18

List the components of cookie technology. 

- file at the local host managed by the user browser
- row in a database of the website for each user
- the server sends a cookie number (id) for the user if it enters the website for the first time

📝 Will be reviewed

19

The ability to inject packets into the Internet with a false source address is known as IP spoofing, while a passive receiver that records a copy of every packet that flies by is called a packet sniffer 

☒ True

☐ False

 **Will be reviewed**

20

Consider that there are 20 servers, and 20 clients connected to the core of the computer network. Assume that there are 20 simultaneous downloads taking place, involving 20 client-server pairs. Suppose that these 20 downloads are the only traffic in the network at the current time. Consider that there is a common link with transmission rate of 5Mbps in the core traversed by all 20 downloads.

Suppose that every server access link has rate of 2 Mbps; every client access link has rate of 1Mbps ; the transmission rates of other links in the core except the one common link are much larger than 5 Mbps. What is the end-to-end throughput for each download? (Answer with justification). 

it will be the smallest rate which is $(5/20) = 0.25\text{Mbps}$

 **Will be reviewed**

21

In packet switching, end-end resources along path are reserved for the duration of communication session 

☐ True

☒ False

 **Will be reviewed**

Given that the requested information is not available at any intermediate databases, a recursive DNS query from a requesting host would follow the path..... 

- ☐ root DNS server, TLD server, local DNS server, authoritative DNS server
- ☐ authoritative DNS server, root server, TLD server, local DNS server
- ☒ local DNS server, root DNS server, local DNS server, TLD server, local DNS server, authoritative DNS server
- ☐ local DNS server, root DNS server, TLD server, authoritative DNS server
- ☐ None of the above

 **Will be reviewed**

Describe two advantages and two challenges of peer-to-peer application architecture. 

advantages is scalability and distribute the traffic (not to be on one server)
 challenges of peer-to-peer is security and the difference between the upload and download rates (upload rates are lower)

 **Will be reviewed**

Consider a sender can transmit at a rate of 1Mbps bits/sec in both directions. Suppose packets containing only control (e.g., ACK or handshaking or request) are 100 bits long. Consider the HTTP protocol, and suppose that the initial downloaded object contains 3 referenced objects from the same server. Suppose that each downloaded object is 100 Kbits long except the last one

which has 200Kbits. Let RTT denotes the round trip time between the local host and the server containing the objects. Find with justification the total time needed to download the objects assuming parallel persistent HTTP. 

you will need three RTT (two for the three way handshaking and one for sending the file)

1) rtt to request the connection $2 \times (100/1 \times 10^6) = 0.0002 \text{ sec}$

2) rtt to request the html file $(100/1 \times 10^6) + ((100 \times 10^3)/1 \times 10^6) = 0.1001 \text{ sec}$

note : (the connection is parallel the the rate will be $(1 \times 10^6)/3$)

3) rtt to request the objects $2 \times ((100 \times 10^3)/((1 \times 10^6)/3)) +$

$((300 \times 10^3)/1 \times 10^6) \times 3 = 1.0003 \text{ sec}$

 **Will be reviewed**

25

Consider that there is a shared link of 1 Mb/s. In case of using TDM with 10 slots per frame and each slot 1000 bit, then the number of frames per second will be 100000. 

☐ True

☒ False

 **Will be reviewed**

26

When the server side receives a command for a file transfer over the control connection (either to, or from, the remote host), the server side initiates a TCP data connection to the client side, then FTP sends exactly one file over the data connection, and then closes the data connection. 

☒ True

☐ False

📝 Will be reviewed

27

Because many firewalls are configured to block (most types of) UDP traffic, designers have increasingly chosen to run multimedia and real-time applications over TCP. 📎

☒ True

☐ False

📝 Will be reviewed

28

Consider that there are two hosts (A & B) connected by 2 packet-switches and three links; every link has transmission rate of R bps. Consider that host (A) sends 4 packets to host (B) back-to-back; each packet has length of L bits. Let d_i and s_i , for $i = 1, 2, 3$, denote the length and the propagation speed respectively of link i starting from host(A). Ignore the processing delay and find with justification the time that the 4 packets take to reach Host (B) in case of one of the packet-switches uses store and forward and the other one uses cut-through. 📎

ps = propagation speed

the time for the packet 1 = $2*(L/R) + ps_1 + ps_2 + ps_3$

the time for the packet the rest = $3*(L/R)$

total = $2*(L/R) + ps_1 + ps_2 + ps_3 + 3*(L/R)$